

Junhyeong Jeon

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Education

Korea Advanced Institute of Science and Technology (KAIST) , Daejeon, South Korea	March 2020 – Present
• Degree: B.S. in Computer Science (School of Computing) • Double Major: Electrical Engineering • Minor: Science and Technology Policy • GPA: 3.83 / 4.3 (Earned 185 Credits)	
Hong Kong University of Science and Technology (HKUST) , Hong Kong	Jun 2023 – Aug 2023
• International Summer Exchange Program	
Yonsei University , Seoul, South Korea	Sep 2022 – Dec 2022
• Academic Exchange Student (Domestic)	
Chungnam National University , Daejeon, South Korea	Dec 2023 - Jun 2024
• Academic Exchange Student (Domestic)	

Research Experiences

mmWave-Based Weapon Detection System (Compressive Sensing)	Sep 2025 – Dec 2025
Affiliation: CyPhy Lab, KAIST (Prof. Jun Han)	
Role: Undergraduate Research Intern	
• Developing a weapon detection system using mmWave with metasurfaces. • Optimized Frequency Selective Surface (FSS) unit-cells to enhance sensing efficiency.	
Effects of Medical School Quota Policy on STEM Human Resources	Jan 2024 – Jun 2024
Affiliation: KAIST URP Program	
Role: First Author (Junhyeong Jeon, Jinwoo Park, Junho Kwon, Soyoung Kim)	
• Conducted quantitative and qualitative analysis on the impact of policy changes on STEM talent supply.	
Exploration of Emerging International Security Issues	Jul 2025 – Oct 2025
Affiliation: FINST&P, KAIST (Commissioned by MOFA)	
Role: Participating Researcher	
• Analyzed security implications of AI and Cybersecurity; compared tech policies of US, China, and EU.	

Work Experience

Batonus , Daejeon, South Korea	Dec 2022 – Feb 2023
Role: Project Manager & AI Developer	
• Led the planning and management of AI-driven projects. • Developed and implemented AI modules for service applications.	

Publications & Presentations

Performance Improvement of Scene Text Detection Using Randomized Preprocessing and Efficient Backbone Utilization	2022
<i>Junhyeong Jeon</i> , Seonghun Park, Seoyoung Park, Hojun Park, Wooseok Jeong, Okjoo Choi (<i>KCC 2022</i>)	
Effects of Medical School Quota Policy on STEM Human Resources	2025
<i>Junhyeong Jeon</i> <i>National STS-STP Undergraduate Conference 2025</i> (Oral Presentation)	
Examining the Sewol Ferry Disaster from the Bottom Up	2022
<i>Junhyeong Jeon</i> , Yeonsu Park <i>Disaster School, KAIST Graduate School of Science and Technology Policy</i> (Presentation)	

Projects

Zero Factory: Integrated Zero-Waste Service Platform	2025
Role: Project Manager & Tech Lead	
Stack: Python (PyTorch, FastAPI), Kotlin, Docker, Kubernetes (EKS), GitHub Actions	
• Led the cross-functional team and defined the technical roadmap for a map-based zero-waste platform. • Developed an AI vision system using ResNet and YOLO to classify reusable vs. disposable containers. • Architected a scalable infrastructure and automated CI/CD pipelines using Docker Compose and GitHub Actions.	

Teaching Experience

Information Security for AI (SW Academy, KAIST)	Mar 2025 – Jun 2025
Main Instructor	
KU Camp Middle School Mentoring (KU Networks)	Aug 2024
Mentor	
FLOW Camp Coding Mentor (Center for Gifted Education, KAIST)	Feb 2024 & Feb 2025
Coding Mentor	
CS101 Introduction to Programming (School of Computing, KAIST)	Sep 2021 – Dec 2024
Teaching Assistant	

Volunteering & Service

Special Lectures for High School Qualification Exam	Sep 2023 – Present
Daejeon Seo-gu Out-of-School Youth Support Center Teaching Mentor	
Informatics Subject Educational Volunteer	Sep 2021 – Aug 2022
Jeolla High School Teaching Volunteer	

Extracurricular Activities & Leadership

Daejeon Youth Policy Network	Mar 2024 – Dec 2024
Role: Team Leader (Culture Team 3)	
Hosted by Daejeon Youth Tomorrow Center & Daejeon Metropolitan City.	
Music Crew PlanB (KAIST & CNU Joint Band Club)	Dec 2020 – Aug 2023
Role: Founder & Representative (2020.12–2022.07)	
Founded a joint university band club and led initial organization and performances.	
Sixline (Acoustic Band) & Eunmak (Indie Film Club)	Sep 2020 – Aug 2022
Active Member, KAIST Undergraduate Clubs	

Honors & Awards

KAIST Academic Excellence Scholarship	Aug 2021 – Aug 2025
Awarded for outstanding academic performance.	
Excellent Volunteer Designation & Bronze Badge	2025
Yuseong-gu Volunteer Center	
Best Mentor Award	2023, 2024
Daejeon Seo-gu Out-of-School Youth Support Center	

Skills

Programming: Python, C/C++, Verilog, SQL, Scala	
Tools & Frameworks: PyTorch, Docker, Kubernetes (EKS), AWS, GitHub Actions	
Languages: Korean (Native), English (Fluent), Chinese (Basic), Japanese (Basic)	